

Zero-Shot Time-Series Foundation Models for Infrastructure Anomaly Detection

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Abstract

Infrastructure monitoring is commonly built from metric-specific rules. Each new signal requires an engineer to select a threshold, a persistence period, and an alert policy. This report asks whether one pretrained time-series foundation model (TSFM) can provide a reusable first anomaly signal across different infrastructure metrics. We evaluate a frozen Toto 2.0 313M model on two operational settings: Ceph storage daemons and a DigitalPath frame-ingestion pipeline. The study separates two questions that initially appeared similar: predicting which storage daemon will fail in the future, and detecting a disturbance while it is occurring. Toto forecasted normal latency well, but its residuals did not produce a practical predictor of genuine unplanned OSD failure. Concurrent detection was more promising when direct operational signals were available: the detector surfaced all 39 scoreable OSD flap proxies and 32 of 34 DigitalPath frame-drop proxies. These high-recall results came with substantial alert volume and did not identify event cause. The evidence therefore supports TSFM as a candidate first-stage disturbance detector, not as a stand-alone failure oracle, root-cause classifier, or replacement for final alert policy.

Keywords: time-series foundation model, anomaly detection, infrastructure monitoring, Ceph, DigitalPath, zero-shot learning

1 Introduction

The operational problem that motivated this work is simple to state. A storage cluster and a data-ingestion pipeline expose many time series, but their alerts are usually designed one metric at a time. A latency metric, a binary up/down state, and a frame counter each have a different scale and event shape. An engineer must learn those differences before choosing a threshold and deciding how long a violation must persist. This process is transparent, but it is slow to extend and difficult to maintain as systems evolve.

A time-series foundation model offers a possible alternative. Instead of encoding one fixed definition of “bad,” it forecasts what a signal should look like next from its recent history. A large difference between the forecast and the observation becomes evidence that the current behavior is unusual. If the same pretrained model can generate useful scores for heterogeneous signals, it could serve as a common first layer while operational rules remain responsible for interpretation and paging.

The project began with the more ambitious goal of predicting storage failures before they occurred. The experiments showed that this goal must be separated from detecting a disturbance that is already underway. They also showed that detecting a disturbance is different from identifying its cause. The report is therefore organized around three questions:

1. **Prediction:** Can current metrics identify the OSD that will experience a genuine unplanned failure within the next 21 days?

2. **Detection:** Can a shared TSFM pipeline surface an OSD state disturbance or DigitalPath flow reduction while it is occurring?
3. **Interpretation:** Can the anomaly score distinguish reboot-associated disturbances from other state changes?

The central finding is that these questions have different answers. The available signals did not support reliable early prediction. Direct state and flow signals did support high-recall concurrent detection, but the anomaly scores alone did not supply operational semantics. This distinction is the throughline for the study rather than an after-the-fact qualification.

2 Operational Problem and Study Design

2.1 Two monitoring environments

The first environment was a production Ceph cluster containing approximately 976 to 991 object storage daemons (OSDs), depending on the extraction period. An OSD is a storage service responsible for data on one device. Prediction experiments used hourly latency, placement-group, operation-rate, disk, network, host, SMART, and state metrics. Concurrent detection focused on a more direct signal: whether the OSD was up and how many native-resolution state transitions occurred within each hour. A *flap* was a repeated change between up and down states.

The second environment was DigitalPath, a pipeline that receives frames through Redis and writes them through an archiver. The primary observable was saved frame rate, with upstream Redis command and byte rates as related channels. The long-range evaluation covered a frozen 197-day interval from February through August 2026. Because no independent incident catalogue was available, a proxy frame-drop event was defined as four consecutive 15-minute samples below 70% of the trailing seven-day median.

2.2 What counted as success

Table 1 separates the evaluation tasks. This separation is important because a detector can obtain high event recall while still being a poor predictor or a noisy paging policy.

Table 1: Research questions and evaluation units.

Question	Positive outcome	Evaluation unit	Primary measures
Early prediction	Alert on the same OSD before its first event within a 21-day horizon	OSD	Recall, precision, false-alert load
Concurrent detection	Alert overlaps an event while it occurs or within the stated grace period	Event and alert episode	Event recall, alert precision, alerts/day, delay
Event interpretation	Separate reboot-associated from non-reboot disturbances without losing target events	Event or host incident	Classification errors and remaining alert load

Recall answers “what fraction of known events were caught?” Precision answers “what fraction of emitted alert episodes matched those events?” These denominators differ. One event may overlap

several alert episodes, so the number of matched alerts can exceed the number of detected events.

2.3 Label quality was part of the experiment

Two label limitations changed how the results should be interpreted. First, a later audit of 44 Ceph labels separated 13 genuine unplanned failures from 31 planned drains or decommissions. Earlier mixed-label prediction results are therefore reported only as a mechanistic comparison, not as genuine-failure performance. Second, the OSD transition and DigitalPath frame-drop labels were derived from signals also presented to the detector. Their recall measures retrospective agreement with operational proxies, not independent root-cause validation. These caveats are stated before the results because they determine which conclusions the numbers can support.

3 A Shared Model and Alert Pipeline

3.1 Model and detection pipeline

The primary model was `Datadog/Toto-2.0-313m`, a probabilistic multivariate forecasting foundation model [1]. It models history within each channel and can use related channels as context, then returns forecast quantiles from q_{10} through q_{90} . The checkpoint remained frozen and no anomaly-specific prediction head was trained.

Prometheus metrics $\rightarrow$ resampling and missing-value masks $\rightarrow$ rolling Toto context $\rightarrow$ quantile forecast $\rightarrow$ channel residual score $\rightarrow$ temporal smoothing and persistence $\rightarrow$ alert episode

The observed value was compared with the forecast range to create a channel residual score. Ceph used a two-sided score because both increases and decreases could matter. DigitalPath used a lower-tail score because the target was reduced flow. Scores were then averaged or persisted over time and grouped into alert episodes. Exact equations and context lengths are provided in Appendix A.

The same model weights were reused, but the wrapper was not universal. Short OSD state changes required little persistence, whereas sustained DigitalPath drops benefited from temporal averaging. “Zero-shot” therefore refers to frozen model weights, not to zero calibration. Channel selection, smoothing, thresholds, and alert grouping were selected using earlier data before the stated holdout whenever possible.

4 Study 1: Can We Predict an OSD Failure Early?

4.1 Hypothesis and experiment

Early prediction asks which device will fail, not merely whether the cluster is experiencing unusual behavior. The initial hypothesis was that a future failing OSD would develop unusually high and persistent latency relative to its own forecast and to peer OSDs. The experiment therefore had three stages:

1. verify that Toto could forecast normal latency better than a seasonal baseline;
2. compare the strongest TSFM residual policy with a simple fleet-relative latency rank; and
3. audit the event labels to separate genuine unplanned failures from planned drains.

Toto passed the forecast sanity check. Median 24-hour MASE was 0.634 for read latency and 0.602 for write latency; values below 1 indicate better forecasts than the selected seasonal-naive baseline [5]. This matters because it shows that the later prediction failure was not simply an inability to model normal latency.

4.2 Result: forecast accuracy did not become failure discrimination

On the preliminary 19-OSD mixed-label cohort, the strongest TSFM residual and a raw fleet-relative latency rank obtained similar recall, but very different precision.

Table 2: Preliminary OSD-level prediction comparison on mixed labels.

Method	Caught	FP OSDs	Precision	False episodes/day
Best Toto residual	12/19	259	4.4%	4.24
Raw latency fleet rank	11/19	23	32.4%	0.39

The 19-OSD cohort predates the final label audit and is not a genuine-failure confusion matrix.

The raw rank was much more precise. The reason was adaptation. Toto produced a large residual when latency first increased, but the elevated level then entered the rolling context and became the model’s new baseline. The residual preserved a short onset burst while the sustained level, which was more useful for this task, faded from the anomaly score.

The later label audit changed the semantic interpretation of the apparent precursors. Of 44 OSDs originally grouped as flap targets, 13 were genuine unplanned failures and 31 were planned drains or decommissions. A previously designed latency, SMART, and hung-state rule system detected 24 of the 44 mixed labels but 0 of the 13 genuine failures. A looser latency rule found 5 of 13, but also alerted on 558 normal OSDs.

Answer to Study 1. The available signals and tested TSFM residuals did not support an operationally useful early predictor of genuine unplanned OSD failure. A good forecast was not automatically a useful prognosis.

5 Study 2: Can We Detect a Disturbance While It Happens?

The negative prediction result motivated a narrower and more operational question: can the same forecast-to-alert mechanism detect an ongoing disturbance? We tested two event shapes. Ceph flapping is short and discrete; DigitalPath degradation is continuous and often sustained.

5.1 Ceph: surfacing availability disturbances

The Ceph detector used hourly OSD state and the count of native-resolution state transitions. The scoring interval covered 304 days and 991 OSDs. After the initial context requirement, 39 labeled OSD rows were scoreable.

Table 3: Recall-first OSD result under the strict repeated-flap proxy.

Detected	Recall	FP episodes	Precision	FP/day
39/39	100%	919	4.1%	3.02

The detector surfaced all scoreable proxy flaps, but the low precision needs interpretation. Many unmatched alerts were not random numeric spikes; they were real one-time down/up events or restarts that did not meet the narrower repeated-flap definition. The detector had learned the broader concept “availability disturbance,” while the label counted only repeated flapping. Residual magnitude did not reliably encode how many transitions occurred.

The result also depended on receiving a signal closely tied to the label. The proxy was defined from state transitions, and transition count was an input. When state, up, and in channels were removed and the detector used indirect performance metrics, Toto detected 0 of 9 scoreable events near a 0.5 false-alert/day budget. Expert rules detected 5 of the same 9 and a naive fleet rank detected 2. This smaller test was coverage-limited, but it prevents a broader claim that any Ceph metric is sufficient.

5.2 DigitalPath: detecting sustained frame-flow reductions

DigitalPath tested whether the shared mechanism could transfer from binary state changes to continuous flow. The frame-only detector used saved frame rate. The multichannel version also supplied acquisition-side context. An original three-hour, high-threshold configuration favored precision; a recall-first one-hour configuration favored event coverage.

Table 4: DigitalPath detection on 34 proxy frame-drop events.

Detector	Events caught	Recall	Unmatched alerts	Precision	Median delay
Frame-only Toto	4/34	11.8%	26	16.1%	2.12 h
Original multichannel Toto	20/34	58.8%	54	21.7%	0.25 h
Recall-first multichannel Toto	32/34	94.1%	246	14.3%	0.50 h

Multichannel context increased event recall substantially, but the operating point controlled the alert cost. The recall-first detector caught 32 of 34 events with a median delay of 30 minutes and emitted 1.46 alert episodes per day. Lowering the threshold enough to catch all 34 events increased unmatched alerts to 871, reduced precision to 6.2%, and raised the alert rate to 4.72 per day. High recall was therefore possible, but it was not free.

To check whether the result was limited to a combined score, each pipeline stage was also scored independently on 25 saved-frame-impact proxies. The archiver counter was the strongest individual channel, catching 24 of 25 events with 7.9% alert-episode precision. Redis command and byte rates caught 23 and 22 events, respectively. This supported channel attribution, but the labels were still rate-derived proxies rather than operator-confirmed incidents.

Answer to Study 2. When an informative operational channel was available, the shared TSFM wrapper could surface ongoing disturbances with high retrospective recall. The same result did not imply low alert volume, independent incident validation, or successful inference from indirect metrics.

6 What the Results Mean Operationally

6.1 Detection is not diagnosis

An OSD reboot and an abnormal non-reboot flap can both look surprising relative to a stable history. Adding boot-derived uptime as another TSFM channel did not solve this semantic classification

problem. A fixed boot-adjusted residual score retained all 37 scoreable no-reboot targets, but reduced false-positive episodes only from 919 to 845; precision changed from 4.1% to 4.2%. Boot residual alone detected only 1 of 2 scoreable reboot-associated positives, so that estimate was too small for a strong classification claim.

Raw operational metadata was more useful. In a comparable no-reboot policy, joining alerts to the exact raw boot epoch retained all 37 targets and reduced false positives from 921 to 559, increasing precision from 3.9% to 6.2%. This is a deterministic post-classifier, not evidence that the TSFM learned reboot semantics. It is also a policy decision: a reboot-associated event is not automatically normal or harmless.

6.2 Simple baselines remained necessary

The public TSB-AD-M benchmark provided an offline reality check on 180 multivariate series [3]. Table 5 reports mean volume under the precision-recall surface (VUS-PR) [4].

Table 5: Offline TSB-AD-M mechanism check.

Method	Mean VUS-PR
Simple direct baseline	0.2915
Toto forecast residual	0.1368
Toto relational deviation	0.1858
Residual/relational ECDF fusion	0.1792

Relational deviation improved on the Toto residual on average, but the simple direct baseline remained stronger overall. The relational score beat the plain residual on 82 of 180 series and lost to the direct baseline on 128 of 180. Moreover, the offline run truncated high-dimensional inputs and used block scores with look-ahead, so it was not an online deployment test. The benchmark reinforced a practical lesson from Ceph and DigitalPath: a complex reusable score should always be compared with a direct, understandable baseline.

6.3 A practical layered design

The experiments support a layered monitoring design rather than a single score that pages operators directly:

1. **Surface:** use the TSFM residual to identify when and where a signal becomes surprising;
2. **Group and attribute:** combine adjacent scores into events and identify which pipeline stage changed;
3. **Interpret:** add boot epochs, host mappings, maintenance records, and other operational metadata; and
4. **Decide:** apply policy-specific grouping, suppression, investigation, or paging.

The evidence is strongest for the first layer. The model supplied a reusable anomaly interface across very different signals, but the event-specific wrapper and later context layers remained necessary.

7 Limitations and Next Validation

- Production evidence came from one Ceph cluster and one DigitalPath installation, so cross-site transfer remains untested.
- OSD and DigitalPath detection labels were derived from detector-visible signals rather than independent operator-confirmed incidents.
- Several OSD rows belonged to the same host-level event and were not statistically independent.
- Some configurations were selected after inspecting the same period; their operating points require an untouched prospective holdout.
- Boot telemetry was missing around some reboot incidents, and reboot-associated does not itself define normality.
- The public benchmark was an offline mechanism test with dimensional truncation and block look-ahead, not a causal online alert evaluation.

The next evaluation should use an operator-confirmed incident catalogue, group correlated OSD rows at host level, freeze all thresholds before the test period, and compare TSFM, PromQL, and direct statistical baselines at the same alert budget. This design would answer the practical question left unresolved by the retrospective studies: whether the TSFM adds useful coverage beyond simpler rules without increasing investigation load beyond an acceptable level.

8 Conclusion

The original goal was to reduce the need for a custom anomaly model or rule for every infrastructure metric. The experiments produced a qualified answer. Toto could forecast normal behavior and reuse the same forecast-to-alert pattern across storage state and frame-flow signals. It did not reliably identify which OSD would experience a genuine unplanned failure, and it did not infer event cause from anomaly magnitude alone.

Concurrent detection was the stronger use case. With direct operational channels, the detector surfaced all 39 scoreable OSD flap proxies and 94.1% of 34 DigitalPath frame-drop proxies at the recall-first operating point. Those results came with 3.02 false-positive episodes per day for the strict OSD label and 1.46 alert episodes per day for DigitalPath. High recall therefore needs to be judged together with alert cost and label independence.

The defensible conclusion is not that TSFM replaces PromQL or diagnoses failures automatically. It is that zero-shot TSFM residuals are a candidate first-stage disturbance signal when informative channels are available. Temporal grouping, raw metadata, simple baselines, and policy-specific rules remain necessary before an operator is paged.

A Technical Scoring Details

Toto produced forecast quantiles for each channel. Most Ceph experiments used the two-sided, unbounded quantile-exceedance score

$$s_t^{\text{two}} = \max \left\{ 0, \frac{y_t - q_{90,t}}{\max(q_{90,t} - q_{50,t}, \epsilon)}, \frac{q_{10,t} - y_t}{\max(q_{50,t} - q_{10,t}, \epsilon)} \right\}. \quad (1)$$

DigitalPath used a clipped lower-tail score because only flow reductions were of interest:

$$s_t^{\text{low}} = \text{clip} \left(\frac{q_{50,t} - y_t}{\max(q_{50,t} - q_{10,t}, \epsilon)}, 0, 1 \right). \quad (2)$$

The exploratory boot-adjusted score normalized and subtracted the boot residual from the transition residual:

$$s_{\text{abnormal}} = \text{clip} \left(\text{clip} \left(\frac{s_{\text{flip}}}{2 \times 10^6}, 0, 1 \right) - \text{clip} \left(\frac{s_{\text{boot}}}{3590.5483}, 0, 1 \right), 0, 1 \right). \quad (3)$$

Prediction sweeps used hourly Ceph metrics, 32–160-hour contexts, 24-hour forecasts, and a 21-day event horizon. Direct OSD detection used a 336-hour context and 24-hour forecast blocks. DigitalPath used 15-minute samples, a 96-hour context, and 32-step forecasts. Missing OSD states were masked. DigitalPath windows required at least 90% observed context per channel.

B Additional Sensitivity Checks

On a separate 119-OSD mixed-label comparison, Toto obtained VUS-PR 0.044 and TimeRCD obtained 0.026 [2]. Both overlapped all 25 mixed flap episodes, but neither provided useful OSD-specific discrimination. The tested full-span TimeRCD configuration used future observations in normalization and context, so its apparent lead time is non-causal and is not treated as a deployable result.

For DigitalPath pipeline-stage attribution, univariate recall-first scoring caught 24 of 25 saved-frame-impact proxies on the archiver counter, 23 of 25 on Redis SET rate, and 22 of 25 on Redis input-byte rate. Corresponding alert-episode precisions were 7.9%, 7.4%, and 6.9%. These values support stage-level disturbance surfacing but remain proxy-label results.

References

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